CPU and decode

File: cpu.kicad_sch

Memory

File: memory.kicad_sch

I/O

File: io.kicad_sch

Power

File: power.kicad_sch

Expansion and debug

File: expansion.kicad_sch

Memory map (24-bit, bank:offset)
  $00:0000–$00:5FFF   RAM (U5)
  $00:6000–$00:7FFF   I/O window, U6 '138, 8 x 1 KB slots
                      $6000 VIA (U7), $7000 ACIA (U8)
  $00:8000–$00:FFFF   ROM (U4)
  $01:0000–$07:FFFF   RAM (U5)
  $08:0000–$7F:FFFF   aliases of banks $00–$07
  $80:0000–$FF:FFFF   VRAMCE decoded; video RAM not fitted (future expansion)

Naming: active-low signals are drawn with an overbar, e.g. $\overline{\text{IOSEL}}$, $\overline{\text{RAMCE}}$, R/$\overline{\text{W}}$.
The W65C datasheets write these with a B suffix (IRQB, RESB, RWB, CS2B);
decode.pld writes them with a leading '/' in its equations.

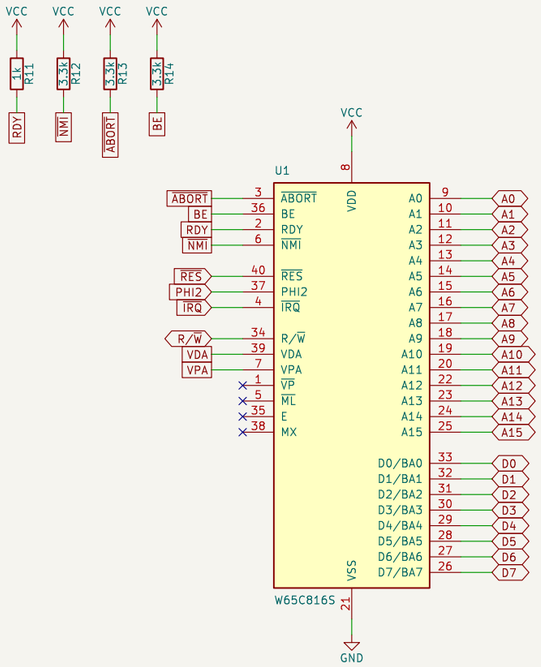
Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

Sheet: /
File: be6502.kicad_sch

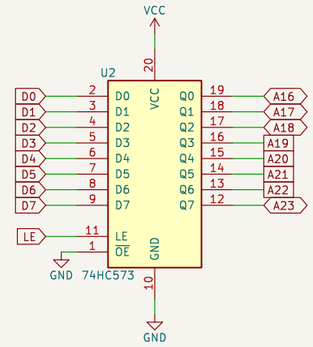
Title: be6502 — W65C816S breadboard computer

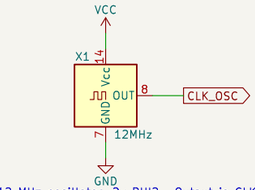
Size: A4	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 1/6

W65C816S (65C816 in DIP-40).
D0-D7 carry the bank byte while PHI2 is low;
U2 captures it while LE is high (see NOTE 6 of decode.pld).
RDY (R11, 1k), NMI, ABORT and BE (R12-R14, 3.3k) have pull-ups
and go to the expansion header. RDY must not be tied straight to +5V: WAI drives it low.
VDA and VPA go to U3 (I/O select qualifier) and the expansion header.
VP, ML, E and MX are not connected.

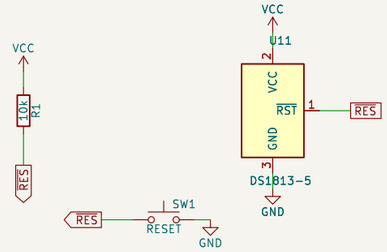


Bank byte latch. Q3-Q6 (A19-A22) go only to the expansion
header: the GAL does not decode them, so banks $08-$7F
alias $00-$07 and $81-$FF alias $80 (decode.pld NOTE 3).





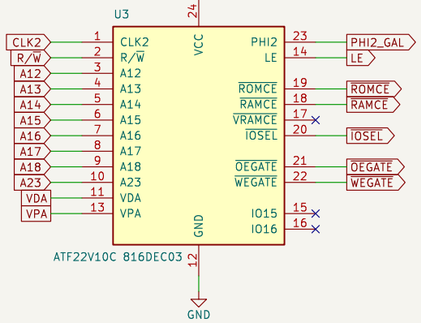
12 MHz oscillator, 2x PHI2. Output is CLK_OSC; JP1 (expansion
sheet) selects it or an external clock as CLK2. U3 divides CLK2
by two to make PHI2 = 6 MHz. Pin 1 not connected.



U11 DS1813-5 reset supervisor: holds RES low ~150 ms after power-up
and debounces SW1 (it stretches any low on RES into a full reset).
RSI is open drain with an internal 5.5k pull-up. R1 (10k) only keeps
RES defined if U11 is left out. -5 suffix = 5% supply tolerance;
use a DS1813-10 if a weak USB supply causes spurious resets.



IRQ is shared by U7 (VIA), U8 (ACIA) and the expansion
header, all open drain, with pull-up R2.
R2: 1k.



U3 runs pld/decode.pld (816DEC03):
```
  PHI2.R    = /PHI2                 (CLK2 / 2)
  LE        = /PHI2 * /CLK2
  /ROMCE    = /A23*/A18*/A17*/A16*A15
  /RAMCE    = bank 0 $0000-$5FFF, or /A23 * (A16+A17+A18)
  /VRAMCE   = A23
  /IOSEL    = /A23*/A18*/A17*/A16*/A15*A14*A13 * (VDA + VPA)
  /OEGATE   = PHI2 * RWB
  /WEGATE   = PHI2 * /RWB
```
VRAMCE (pin 17) is not connected: no video RAM this generation.
Pin 23 drives PHI2_GAL; JP2 (expansion sheet) picks direct or buffered.
A12 is wired but unused (NOTE 4). Pins 15, 16 spare.
Burn with power-down disabled (NOTE 5).

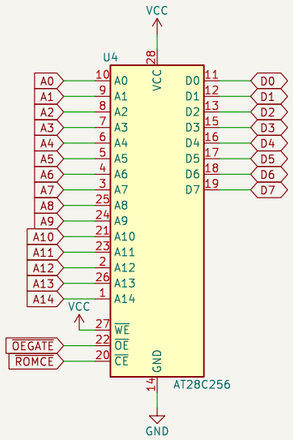
Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer - see README.md and pld/decode.pld
be6502

Sheet: /CPU and decode/
File: cpu.kicad_sch

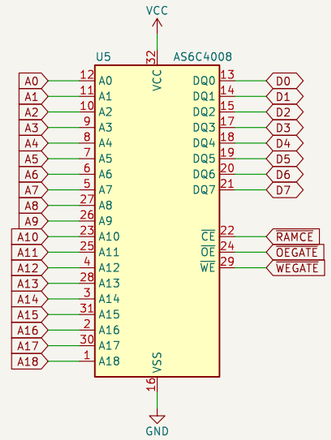
Title: CPU, clock, bank latch and address decode

Size: A3	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 2/6

ROM $00:8000–$00:FFFF (32 KB)
WE tied high: not writable in circuit.



RAM 512 KB. A16–A18 come from the bank latch (U2).
$00:0000–$00:5FFF and $01:0000–$07:FFFF reachable.



OEGATE is shared by the ROM and RAM; their chip enables are mutually exclusive, so at most one device drives D0–D7 (decode.pld NOTE 1).
WEGATE goes only to the RAM. It includes PHI2, so no write can happen while D0–D7 carries the bank byte.
Video RAM (62256 on VRAMCE, bank $80) is reserved for future expansion.

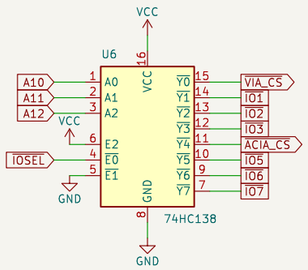
Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

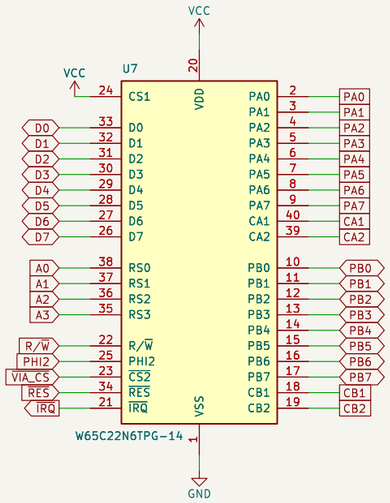
Sheet: /Memory/
File: memory.kicad_sch

Title: Memory: ROM and RAM

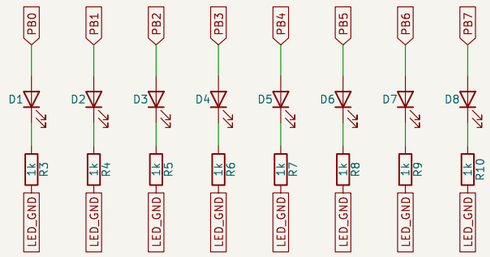
Size: A3	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 3/6

I/O window $00:6000–$00:7FFF, 8 x 1 KB slots on A12:A11:A10
  Y0 $6000 VIA Y4 $7000 ACIA
  Y1 $6400 Y2 $6800 Y3 $6C00
  Y5 $7400 Y6 $7800 Y7 $7C00 (to the expansion header as IOn)
E1 (pin 5) is the reserved PHI2 hook: tie to PHI2
to qualify the selects later (decode.pld NOTE 2).
E2 (pin 6) is the active-high enable — it must be high.

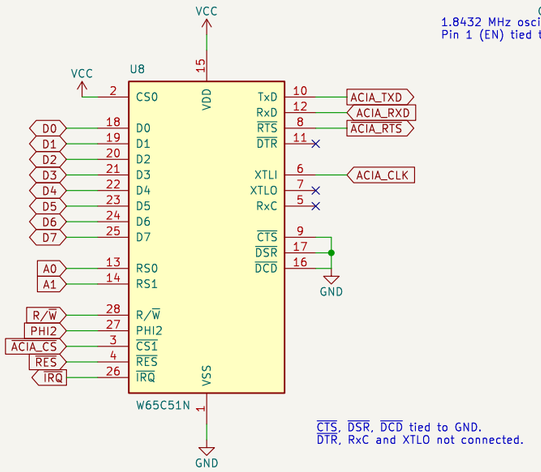




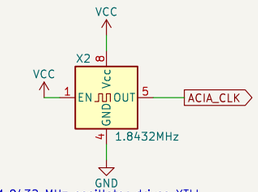
Port A, Port B, CA1, CA2, CB1 and CB2 go to the
VIA header J4 (expansion sheet).
IRQB is open_drain on the W65C22N: wired
straight to IRQ (pull-up R2, CPU sheet).



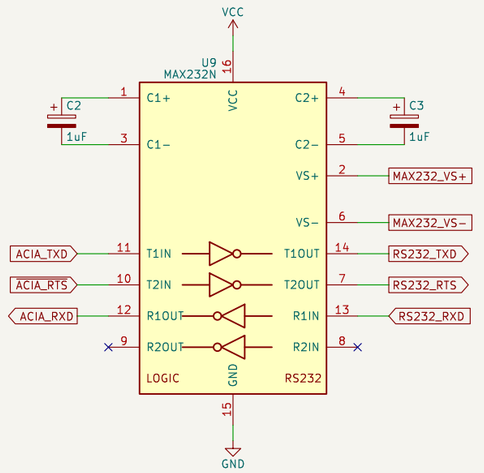
Port B LEDs: PBn -> LED -> resistor -> LED_GND.
LED_GND reaches GND through JP3 (expansion sheet);
pull the jumper to free Port B for the header. R3–R10: 1k.



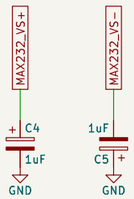
CTS, DSR, DCD tied to GND.
DTR, RxC and XTLO not connected.

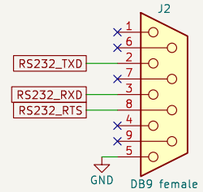


1.8432 MHz oscillator drives XTLI.
Pin 1 (EN) tied to +5V.



MAX232N: T1 = ACIA TxD, R1 = ACIA RxD, T2 = ACIA RTS.
R2IN and R2OUT not connected. C2–C5 1uF electrolytic;
C4 from VS+ to GND, C5 from VS- to GND.





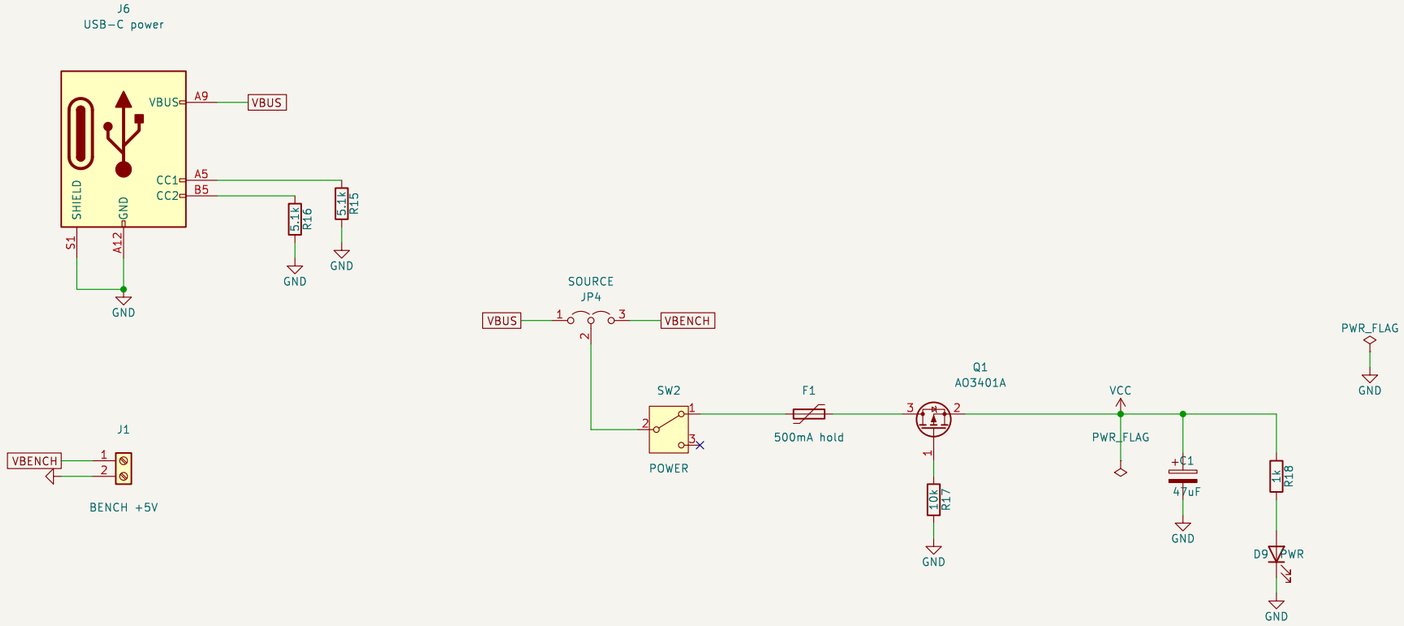
J2 DB9 female: 2 TxD (T1OUT), 3 RxD (R1IN),
8 RTS from ACIA (T2OUT), 5 GND.
bios.s: 115200 8-N-1, flow control on ACIA RTS.

Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
be6502

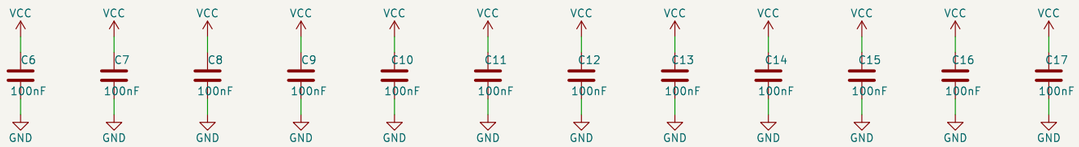
Sheet: /I/O/
File: io.kicad_sch

Title: I/O: '138 slot decoder, VIA, ACIA, RS-232

Size: A3	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 4/6



Power input. J6 USB-C (power only: CC1/CC2 5.1k to GND = default USB 5 V)
or J1 bench supply. JP4 picks ONE source: 1-2 = USB, 2-3 = bench.
Never connect both with the jumper set to one - the other is simply unused.
SW2 power switch -> F1 0.5 A polyfuse -> Q1 reverse-polarity P-FET
(drain to input, source to +5V, gate to GND via R17). Q1 blocks a
reversed bench supply. C1 47uF bulk. D9 power LED (R18 1k).



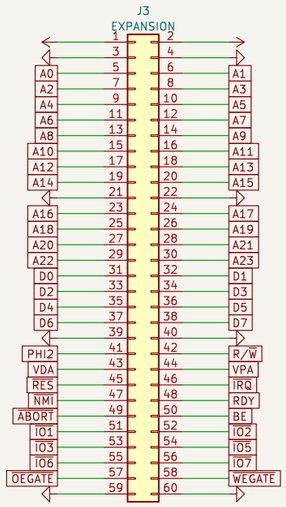
Decoupling: 100nF ceramic, one per package, placed at its VCC pin.
C6 U1 W65C816S
C7 U2 74HC573
C8 U3 ATF22V10C
C9 U4 AT28C256
C10 U5 AS6C4008
C11 U6 74HC138
C12 U7 W65C22
C13 U8 W65C51N
C14 U9 MAX232
C15 U10 74ACT08
C16 X1 12MHz
C17 X2 1.8432MHz

Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S computer - see README.md and pld/decode.pld
be6502

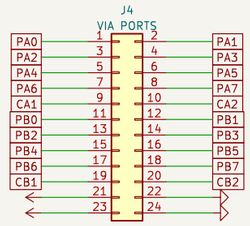
Sheet: /Power/
File: power.kicad_sch

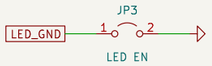
Title: Power: input, protection, decoupling

Size: A3	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 5/6

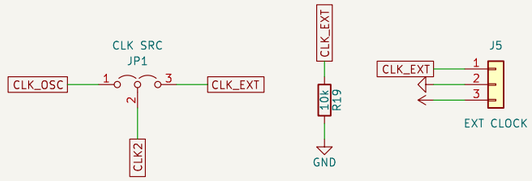


J3 expansion bus (2x30 IDC, keyed).
A16–A23 are the latched bank byte (U2), valid while PHI2 is high.
A19–A22 are not decoded on the board, so an expansion device in banks $08–$7F collides with the on-board RAM aliases: use the IOn slots ($6400–$7FFF, 1 KB each) for expansion I/O.
IRQ, NMI, RDY, ABORT, BE have pull-ups on the CPU sheet: drive them open drain only.

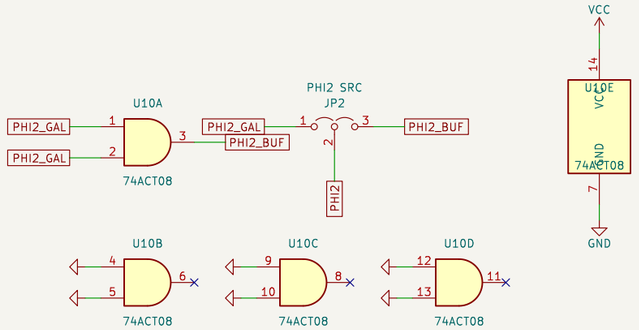




J4 VIA header. Port B also drives the LEDs D1–D8; remove JP3 to disconnect them.



Clock source. JP1 1–2: X1 oscillator (normal). JP1 2–3: J5 external clock, e.g. a single-step or 555 clock module.
R19 keeps CLK_EXT low when nothing is plugged in.
The GAL divides CLK2 by two: two external pulses per bus cycle.



PHI2 drive. The ATF22V10C only guarantees VOH = 2.4 V at 4 mA; the W65C816S wants about 3.5 V on PHI2. JP2 1–2 (default): PHI2 straight from GAL pin 23, as on the breadboard. JP2 2–3: through U10A, a TTL-input (VIH 2.0 V) gate with full-swing outputs. The buffer delays PHI2 by up to ~10 ns relative to the GAL's internal PHI2; see decode.pld NOTE 8. Unused gates: inputs to GND.



Test points: +5V, two GND loops for scope grounds, PHI2, CLK2, RES, IRQ. The full bus is also on J3 for a logic analyzer or an Arduino bus monitor.



Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S computer — see README.md and pld/decode.pld
be6502

Sheet: /Expansion and debug/
File: expansion.kicad_sch

Title: Expansion headers and debug aids

Size: A3	Date: 2026-10-16	**Rev: 2**
KiCad E.D.A. 9.0.4		Id: 6/6